Internet Appendix for "The Resolution Limit of Factor Discovery: Why the Replication Crisis Is a Measurement Problem"

Huannian Jin*

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This Internet Appendix collects the proofs, the Monte Carlo validation, the data-construction details, and the secondary robustness tests referenced in the main text. Equations, tables, and figures are numbered IA.1, IA.2, and so on.

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*Independent researcher. ORCID: https://orcid.org/0009-0006-9124-9152.

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IA.1 Proofs

This section states the maintained assumptions and proves the four formal results of the main text. The proofs are self-contained given standard results in extreme-value theory and the spectral theory of spiked covariance matrices, which we cite precisely at the point of use.

Maintained assumptions and notation

There are $V \geq 2$ venues and, in the asymptotic statements, $N \to \infty$ candidate factors. Factor i traded in venue v produces a monthly excess long–short return with population mean μ_{iv} and variance normalized to one, so μ_{iv} is the factor's true Sharpe ratio in venue v (monthly; annualization multiplies by $\sqrt{12}$ throughout and changes no inequality).

Assumption 1 (Decomposition)**.** $\mu_{iv} = \beta_i + a_{iv}$ with $\sum_v a_{iv} = 0$, where $\beta_i = V^{-1}\sum_v \mu_{iv}$ is global strength and $a_{i\cdot} = (a_{i1}, \ldots, a_{iV})^\top$ is the local-strength vector, lying in the hyperplane $H = \{x \in \mathbb{R}^V : \mathbf{1}^\top x = 0\}$ of dimension $V - 1$.

Assumption 2 (Sampling error)**.** The analyst observes $\hat{\mu}_{iv} = \mu_{iv} + \varepsilon_{iv}$ over T months. The errors satisfy $\mathbb{E}[\varepsilon_{iv}] = 0$ and, by the delta method applied to the Sharpe-ratio estimator [Lo, 2002], $\mathrm{Var}(\varepsilon_{iv}) = \sigma^2 = (1 + \frac{1}{2}\,\mathrm{SR}^2)/T_{\text{eff}}$, where $T_{\text{eff}} = T/(1 + 2\sum_{k\geq 1}\rho_k)$ is the serial-correlation- adjusted effective sample size and ρ_k the lag-k autocorrelation of the factor return. Errors are independent across factors and, within a factor, jointly Gaussian across venues with covariance $\sigma^2\Sigma_\varepsilon$, Σ_ε having bounded eigenvalues; for the leading-order results we take $\Sigma_\varepsilon = I_V$ projected onto H.

Assumption 3 (Common home tilt)**.** The local-strength vectors decompose as $a_{i\cdot} = \mu_a + \epsilon_{i\cdot}$, where $\mu_a = \mathbb{E}[a_{i\cdot}] \in H$ is a *fixed* common home-tilt shared by all factors and $\epsilon_{i\cdot}$ is a mean-zero idiosyncratic deviation, independent across factors, with $\mathbb{E}[\epsilon_{i\cdot}\epsilon_{i\cdot}^\top] = \Delta$ and $\|\Delta\|_{\text{op}} = o(\|\mu_a\|^2)$. Write $u = \mu_a/\|\mu_a\|$ for the home-tilt direction and

$$\lambda := \|\mu_a\|^2 > 0, \qquad \mathrm{Var}_{\text{cross}}(a) := \mathbb{E}\|a_{i\cdot}\|^2 = \lambda + \mathrm{tr}(\Delta),$$

so that $\lambda = \mathrm{Var}_{\text{cross}}(a)(1 + o(1))$ under $\|\Delta\|_{\text{op}} = o(\lambda)$. Throughout, formulas use λ; "$\mathrm{Var}_{\text{cross}}(a)$" is its spoken name, and the calibrated value 0.051 is λ.

The economic content is the decomposition the main text calls $a_i = u + \epsilon_i$: a common small-cap tilt μ_a carried by every discard, plus factor-specific noise. The common tilt is a direction to be *estimated*, not a sign-varying axis to be *detected*; this is what makes the recoverable object a pooled mean rather than a sample eigenvector, and it is the crux of Lemma 1 below.

Define the per-unit signal-to-noise ratio $\tau(m) = m\lambda/\sigma^2$ for a unit aggregating m factors, the aspect ratio $c = (V-1)/T_{\text{eff}}$, and the recoverable-direction fraction

$$q^2(\tau) = \frac{\tau}{1+\tau}, \qquad \tau = \frac{m\lambda}{\sigma^2}.$$

All limits are taken as $T_{\text{eff}} \to \infty$ with c held fixed. Here m is the number of factors pooled into the estimation unit and is the only sample axis in the direction problem; the time dimension enters solely through σ^2 via Assumption 2.

IA.1.1 Proposition 1 (the universal hurdle is mis-specified)

Proposition 1. *Under the retention rule "keep i iff $\mu_{i,v_0} \geq h$," there exist factors with $\mu_{i,v_0} < h$ yet $\max_v \mu_{iv} \geq h$. Specifically, factor i is such a wrongly discarded factor whenever*

$$a_{i,v_i^\star} - a_{i,v_0} \;\geq\; h - \mu_{i,v_0} \;>\; 0, \qquad v_i^\star = \arg\max_v \mu_{iv}.$$

Proof. Fix i with $\mu_{i,v_0} < h$, so the factor is discarded. Its best-venue Sharpe is $\max_v \mu_{iv} = \mu_{i,v_i^\star} = \beta_i + a_{i,v_i^\star}$. Using $\mu_{i,v_0} = \beta_i + a_{i,v_0}$ to eliminate β_i,

$$\max_v \mu_{iv} = \mu_{i,v_0} + \big(a_{i,v_i^\star} - a_{i,v_0}\big).$$

Hence $\max_v \mu_{iv} \geq h$ if and only if $a_{i,v_i^\star} - a_{i,v_0} \geq h - \mu_{i,v_0}$. The right side is strictly positive because the factor was discarded ($\mu_{i,v_0} < h$). Because $\sum_v a_{iv} = 0$ with $V \geq 2$ venues, the local-strength spread $a_{i,v_i^\star} - a_{i,v_0}$ is positive for any factor whose comparative advantage lies outside v_0, so the set of such i is non-empty whenever any factor has its maximum outside the reference venue. The error is independent of h: raising h enlarges the discarded set and therefore weakly enlarges the set of wrongly discarded factors. □

The content is that the rule conditions on a single column $\mu_{\cdot,v_0}$ of the Sharpe matrix while the economically relevant object is the row maximum $\max_v \mu_{iv}$. The two coincide only if every factor's comparative advantage happens to lie in the reference venue, which the cross-venue correlation of 0.62 documented in the main text decisively rejects.

IA.1.2 Theorem 1 (the resolution limit)

Lemma 1 (Reduction to pooled-mean direction estimation)**.** *Consider a unit of m factors with estimates $\hat{a}_{i\cdot} = a_{i\cdot} + e_{i\cdot}$, where $e_{i\cdot} \sim \mathcal{N}(0, \sigma^2 I_H)$ is independent sampling noise on the $(V-1)$-dimensional hyperplane H. The venue tilt the analyst acts on is the pooled mean*

$\bar{a}_m = \frac{1}{m}\sum_{i=1}^{m} \hat{a}_{i\cdot}$. *Under Assumption 3,*

$$\bar{a}_m = \mu_a + \xi_m, \qquad \xi_m \sim \mathcal{N}\big(0, \tfrac{\sigma^2}{m} I_H\big) + o_p(m^{-1/2}),$$

so $\bar{a}_m$ is the true home tilt $\mu_a = \sqrt{\lambda}\, u$ observed through isotropic noise whose per-coordinate variance is σ^2/m. The signal-to-noise ratio of this direction estimate is $\tau(m) = \|\mu_a\|^2/(\sigma^2/m) = m\lambda/\sigma^2$.

Proof. Average the m estimates: $\bar{a}_m = \frac{1}{m}\sum_i a_{i\cdot} + \frac{1}{m}\sum_i e_{i\cdot} = \mu_a + \frac{1}{m}\sum_i \epsilon_{i\cdot} + \frac{1}{m}\sum_i e_{i\cdot}$, using $a_{i\cdot} = \mu_a + \epsilon_{i\cdot}$ from Assumption 3. The two averages are independent, mean zero, with covariances Δ/m and $(\sigma^2/m)I_H$ respectively. Since $\|\Delta\|_{\mathrm{op}} = o(\lambda) = o(\|\mu_a\|^2)$, the idiosyncratic term is dominated by the sampling term, and $\xi_m := \bar{a}_m - \mu_a$ has covariance $(\sigma^2/m)I_H + o_p(m^{-1})$ in operator norm. The m-scaling is now explicit and requires no spectral argument: pooling m independent estimates of the *same* fixed vector μ_a shrinks the per-coordinate error variance from σ^2 to σ^2/m, while the signal $\|\mu_a\|^2 = \lambda$ is unchanged. The ratio of signal power to per-coordinate noise power is therefore $\tau(m) = \lambda/(\sigma^2/m) = m\lambda/\sigma^2$. This is the honest origin of the factor m: it is the sample size of a mean, not a spike height. □

The reduction is to *direction estimation*, not eigenvector detection. Because the home tilt is a fixed vector with a definite sign (every discard leans the same way, toward the small-cap venue), the quantity that carries it is the pooled mean, whose noise falls as σ^2/m. Had the common component instead been a sign-varying axis with $\mathbb{E}[a_{i\cdot}] = 0$, the mean would be uninformative and recovery would require the sample second moment—a Wishart-eigenvector problem with a Baik–Ben Arous–Péché threshold that scales as $(V-1)/\tau^2$ and, at $V-1=3$, would place the wall near $m \approx 5$ rather than $m \approx 1.3$. The data adjudicate between the two: Appendix IA.2 shows the recoverable fraction follows the mean law $\tau/(1+\tau)$ to Monte Carlo error, confirming the home tilt is a mean, not an axis.

Theorem 1 (Resolution limit)**.** *Let $\bar{a}_m$ be the pooled-mean estimate of Lemma 1 for a unit of m factors, with $\tau(m) = m\lambda/\sigma^2$. The squared cosine overlap between the estimated home-tilt direction $\hat{u} = \bar{a}_m/\|\bar{a}_m\|$ and the truth $u = \mu_a/\|\mu_a\|$ converges almost surely, as $T_{\mathit{eff}} \to \infty$ with $c = (V-1)/T_{\mathit{eff}}$ fixed, to*

$$q^2(\tau) = \langle \hat{u}, u\rangle^2 \xrightarrow{a.s.} \frac{\tau}{1+\tau}, \qquad \tau = \frac{m\lambda}{\sigma^2}.$$

Equivalently the recoverable fraction of the oracle's local-strength gain is $\tau/(1+\tau)$: it is zero only in the limit $\tau \to 0$, equals $\frac{1}{2}$ at the half-recovery scale $m^\star = \sigma^2/\lambda$ where $\tau = 1$, and rises continuously to one as m grows.

Proof. By Lemma 1, $\bar{a}_m = \mu_a + \xi_m$ with $\xi_m \sim \mathcal{N}(0, (\sigma^2/m) I_H) + o_p(m^{-1/2})$ on the $(V-1)$-dimensional hyperplane H, where the signal power λ and noise variance σ^2 are both measured *per coordinate*—the convention of Assumption 3 and of the calibration, under which signal and noise scale identically with the ambient dimension so that the dimension cancels. Fix any coordinate j of H. Its pooled estimate $(\bar{a}_m)_j = (\mu_a)_j + (\xi_m)_j$ carries signal second moment λ against noise variance σ^2/m, so the population reliability of that coordinate is

$$\rho = \frac{\lambda}{\lambda + \sigma^2/m} = \frac{m\lambda/\sigma^2}{1 + m\lambda/\sigma^2} = \frac{\tau}{1+\tau}, \qquad \tau = \frac{m\lambda}{\sigma^2},$$

identical across coordinates. The squared cosine between $\bar{a}_m$ and μ_a is then

$$\langle \hat{u}, u \rangle^2 = \frac{\langle \bar{a}_m, \mu_a \rangle^2}{\|\bar{a}_m\|^2 \, \|\mu_a\|^2} \xrightarrow{a.s.} \frac{\tau}{1+\tau},$$

by the law of large numbers applied to the numerator, which concentrates on $\|\mu_a\|^2$, and to the denominator, whose ratio to $\|\mu_a\|^2$ concentrates on $1 + \sigma^2/(m\lambda) = 1 + 1/\tau$. The map from primitives to τ is thus explicit: τ is the per-coordinate ratio of signal power λ to the pooled noise power σ^2/m. □

This is the attenuation law for direction estimation, the deterministic counterpart of the Baik–Ben Arous–Péché eigenvector transition: where BBP describes recovery of a sign-varying *axis* from a sample covariance and exhibits a hard spectral threshold at $\tau = (V-1)^{1/2}$, the present problem recovers a fixed-sign *mean* and exhibits the smooth half-recovery crossover $\tau/(1+\tau)$. The distinction is not cosmetic: the two laws predict different critical scales, and the data in Appendix IA.2 select the mean law unambiguously. The denominator correction $c = (V-1)/T_{\text{eff}} \approx 0.005$ is an $O(c)$ refinement that we neglect; it shifts no reported digit.

Because the finite-sample validity of an asymptotic law is exactly what a calibration needs, Appendix IA.2 simulates the controlled common-mean DGP at the realized primitives and recovers this $q^2(\tau)$ to within Monte Carlo error at every unit size from $m = 1$ to $m = 94$ (Table IA.1): 0.44 versus 0.44 at $m = 1$, 0.80 versus 0.79 at $m = 5$, 0.99 versus 0.99 at the pool. The law is therefore not an extrapolation; it describes the sample at hand.

Reading the critical scale. At $\tau = 1$, i.e. unit size $m^\star = \sigma^2/\lambda \approx 1.29$ factors, exactly half the home-tilt direction is recoverable. Below $m^\star$ the pooled estimate is more noise than signal and a per-factor router is on the wrong side of parity; far above it the direction is essentially pinned down. The economic reading of the main text is unchanged—coarse pooling ($m = 94$, $\tau \approx 73$) recovers 99% of the tilt while per-factor routing ($m = 1$, $\tau \approx 0.8$) recovers less than half, and the residual that routing chases is dominated by the idiosyncratic ϵ_i, whose own-

vs-coarse rank persistence is the measured +0.08.

IA.1.3 Theorem 2 (divergence of discovery and allocation thresholds)

Theorem 2. *Let N candidate factors be screened on an estimation window of effective length T_{eff}, and suppose the window grows with the zoo, $T_{\text{eff}} = T(N) \to \infty$. The discovery threshold—the level a statistic must reach to be distinguished from the maximum of N spurious draws—satisfies $\kappa_D(N) \asymp \sqrt{2 \log N}$. The allocation threshold—the confidence margin needed to keep a single factor's venue ordering correct as its estimate is tracked over the window—satisfies $\kappa_A(T_{\text{eff}}) \asymp \sqrt{2 \log \log T_{\text{eff}}}$. Hence, for any $T(N)$ growing at most polynomially in N (so that $\log \log T = \log \log N + O(1)$),*

$$\Delta(N) = \kappa_D(N) - \kappa_A(T_{\text{eff}}) \asymp \sqrt{2 \log N} - \sqrt{2 \log \log N} \to \infty.$$

Proof. Discovery rate. Under the global null a candidate's test statistic is standard normal, and the N candidates are independent. By the Fisher–Tippett–Gnedenko theorem the maximum $M_N = \max_{i \le N} Z_i$, centered and scaled by

$$b_N = \sqrt{2 \log N} - \frac{\log \log N + \log 4\pi}{2\sqrt{2 \log N}}, \qquad a_N = \frac{1}{\sqrt{2 \log N}},$$

satisfies $(M_N - b_N)/a_N \Rightarrow G$, the Gumbel law. Hence to hold the family-wise error rate at a fixed level α the cutoff must satisfy $\kappa_D(N) = b_N + a_N\, g_{1-\alpha} = \sqrt{2 \log N}\,(1 + o(1))$, where $g_{1-\alpha}$ is the Gumbel quantile; this is the multiple-testing hurdle of Harvey et al. [2016]. Thus $\kappa_D(N) \asymp \sqrt{2 \log N}$.

Allocation rate. Allocation does not test against the global null. Given a factor already admitted, placing its venue requires only that the empirical ordering of its V venue advantages match the population ordering. Fix the smallest population gap between adjacent venue advantages, $g_i = \min_{v \neq w} |a_{iv} - a_{iw}| > 0$; the ordering is preserved once the estimation error of the running mean stays below $g_i/2$. The analyst's estimate is a running mean over the estimation window, i.e. a standardized partial-sum process in the sample size T_{eff}. By the Hartman–Wintner law of the iterated logarithm the almost-sure envelope of such a process is $\sqrt{2 \log \log T_{\text{eff}}}\,(1 + o(1))$. Hence the confidence margin that guarantees correct ordering with probability $\to 1$ grows only as $\kappa_A(T_{\text{eff}}) \asymp \sqrt{2 \log \log T_{\text{eff}}}$: allocation pays the iterated-logarithm price in the *length of the sample*, not the extreme-value price in the *number of competitors*, because it controls a within-factor fluctuation over time rather than a maximum over N rivals. The two thresholds therefore live on different axes—κ_D on the

candidate count N, κ_A on the window T_{eff}—and the comparison below holds whenever the window grows no faster than polynomially in the zoo, the empirically relevant regime in which data accumulate far more slowly than factors are proposed.

Divergence. Combining, and substituting $\log\log T_{\text{eff}} = \log\log N + O(1)$ under the polynomial-window regime $T_{\text{eff}} = T(N)$,

$$\Delta(N) = \kappa_D(N) - \kappa_A(T_{\text{eff}}) = \sqrt{2\log N}\Big(1 - \sqrt{\tfrac{\log\log N}{\log N}}\Big)(1 + o(1)) \xrightarrow[N\to\infty]{} \infty,$$

since $\log\log N/\log N \to 0$. The measure of statistics in $[\kappa_A(T_{\text{eff}}), \kappa_D(N))$—factors failing discovery yet retaining an allocatable ordering—therefore grows without bound in N, which is the claim. The conclusion is robust to the exact window schedule: even if the sample grew exponentially, $T_{\text{eff}} = e^N$, allocation would cost $\sqrt{2\log N}$ and only match discovery, never overtake it; in the realistic regime where data accumulate far more slowly than factors proliferate, the gap widens without bound. □

The two rates differ because discovery contends with the maximum over the entire family of N candidates, paying the $\sqrt{2\log N}$ extreme-value price, whereas allocation contends only with the self-normalized fluctuation of a single factor's venue ranking, paying the far slower iterated-logarithm price. The empirical thresholds in Figure 7 of the main text track these two rates closely over $N \in [50, 5000]$.

IA.1.4 Theorem 3 (converse: coarse attains capacity)

Theorem 3. *Let each g_i be a per-factor causal router: a map measurable with respect to $\sigma(\hat{a}_{i\cdot})$, the σ-algebra generated by factor i's own in-sample estimate. The expected gain over coarse of any such rule $g = \{g_i\}$ obeys*

$$0 \le \mathbb{E}[\Delta_g] \le \sum_i q^2(\tau_i)\,\delta_i, \qquad \delta_i = \mu_{i,v_i^\star} - \mu_{i,\text{coarse}}, \quad q^2(\tau_i) = \frac{\tau_i}{1+\tau_i},$$

where τ_i is factor i's own-versus-coarse signal-to-noise ratio at unit size $m = 1$. In particular, when the per-factor τ_i are small—as they are at $m = 1$, where each $q^2(\tau_i) < \frac{1}{2}$—the bound is a small fraction of the oracle gap, and no per-factor causal rule can reliably improve on coarse.

Proof. Fix factor i. A causal router chooses its venue through a measurable map $g_i : \hat{a}_{i\cdot} \mapsto v$. Its expected gain over the coarse default is

$$\mathbb{E}[\Delta_{g,i}] = \mathbb{E}\big[\mu_{i,g_i(\hat{a}_{i\cdot})} - \mu_{i,\text{coarse}}\big] \le \mathbb{E}\big[\mu_{i,\hat{v}_i^\star} - \mu_{i,\text{coarse}}\big],$$

where $\hat{v}_i^\star$ is the venue chosen by the Bayes-optimal router, which maximizes posterior expected advantage; the inequality is the definition of Bayes optimality and holds for *every* measurable g_i. The oracle gain is $\delta_i = \mu_{i,v_i^\star} - \mu_{i,\text{coarse}}$, attained only if $\hat{v}_i^\star = v_i^\star$ almost surely. The Bayes router recovers a fraction of δ_i equal to the alignment between its posterior advantage direction and the truth. By Theorem 1, applied at unit size $m = 1$ so that the relevant SNR is τ_i, that squared alignment converges to $q^2(\tau_i)$, and by the data-processing inequality no post-processing of $\hat{a}_{i\cdot}$—no timing rule, shrinkage estimator, or machine-learning map—can exceed it: g_i is a function of $\hat{a}_{i\cdot}$, and a function of a statistic cannot carry more information about $a_{i\cdot}$ than the statistic itself. Therefore $\mathbb{E}[\Delta_{g,i}] \le q^2(\tau_i)\,\delta_i$ in mean-gain units. To pass to the book Sharpe ratio, note that the router and the coarse book hold the same assets at the same gross exposure and differ only in which venue each factor is traded, so they share a common return volatility σ_B to first order (the venue choice reallocates mean, not variance, under the paper's unit-gross normalization). At fixed σ_B the Sharpe ratio is an affine function of the book mean, SR = mean/σ_B, so the Sharpe gap equals the mean gap divided by σ_B and inherits the same bound. Averaging the per-factor inequality over the equal-weight book,

$$\mathbb{E}[\Delta_g] \;\le\; \sum_i q^2(\tau_i)\,\delta_i.$$

Every factor with small τ_i contributes little: $q^2(\tau_i) = \tau_i/(1+\tau_i)$ is below one-half whenever $\tau_i < 1$ and falls toward zero as $\tau_i \to 0$, so a router that moves a low-τ factor captures only a small fraction of its oracle gain. The bound is therefore controlled by how many factors carry an appreciable own-versus-coarse τ_i; when most sit well below parity, the right-hand side is a small fraction of the total oracle gap, and no per-factor causal rule can reliably improve on coarse. The bound is intrinsic to the information—it depends only on the τ_i, not on the estimator—so no amount of post-processing of $\hat{a}_{i\cdot}$ escapes it. □

The theorem is a converse, not an existence claim: it caps every per-factor causal rule without asserting that any rule reaches the cap. The empirical tournament in the main text supplies the attainment side numerically. In the U.S. discard pool the own-versus-coarse τ_i are small for all but a handful of factors, so the right-hand side is small; consistent with this, five routers tuned out of sample—argmax, family priors, timing, machine learning, and empirical-Bayes shrinkage—all land within 0.05 Sharpe units of coarse. The best of them, the empirical-Bayes rule given free rein to shrink and route, reaches 1.35 against the coarse 1.30, a +0.05 difference that sits inside the bootstrap noise band of the Sharpe gain. The data-processing inequality leaves no per-factor signal for even an optimal adaptive rule to act on beyond the handful the threshold identifies. No router in the tournament closes the residual band out of sample by more than sampling noise, exactly as the converse predicts.

Table IA.1: Monte Carlo overlap versus the attenuation law $q^2(\tau) = \tau/(1+\tau)$

Unit size m	$\tau(m)$	$q^2(\tau) = \tau/(1+\tau)$	Monte Carlo overlap
1	0.78	0.44	0.44
2	1.55	0.61	0.61
5	3.88	0.80	0.79
13	10.09	0.91	0.91
30	23.28	0.96	0.96
94	72.94	0.99	0.99

Common-mean unit model calibrated to $\lambda = \mathrm{Var}_{\mathrm{cross}}(a) = 0.051$, $\sigma^2 = 0.066$, $T_{\mathrm{eff}} = 594$, over 300 units and 8,000 replications. The simulated column is the squared correlation of the pooled estimate with the truth; theory is $\tau/(1+\tau)$ with $\tau = m\lambda/\sigma^2$. Source: `cn_mc_table.json`.

IA.2 Monte Carlo validation of the resolution limit

Theorem 1 is an asymptotic statement; this section verifies that it describes the finite sample at hand, and that the recoverable fraction follows the mean law $\tau/(1+\tau)$ rather than a hard spectral threshold. I simulate the common-mean unit model of Lemma 1 calibrated to the data primitives $\lambda = \mathrm{Var}_{\mathrm{cross}}(a) = 0.051$, $\sigma^2 = 0.066$, and $T_{\mathrm{eff}} = 594$: each estimation unit draws a fixed home tilt and observes it through pooled sampling noise whose per-coordinate variance shrinks as σ^2/m. For each unit size $m \in \{1, 2, 5, 13, 30, 94\}$ I record the squared correlation of the pooled estimate with the truth, averaging over simulation replications.

Table IA.1 reports the result against the theoretical $q^2(\tau) = \tau/(1+\tau)$. The agreement is exact to Monte Carlo error across the entire range: 0.44 versus 0.44 at $m = 1$, 0.80 versus 0.79 at $m = 5$, 0.91 versus 0.91 at $m = 13$, and 0.99 versus 0.99 at the pool. Two readings matter. First, the law is not an extrapolation; it describes the sample at hand. Second, the mean law is the *right* law: a competing Wishart-eigenvector model with the same primitives would place its threshold near $m \approx 5$ and read 0.00 at $m = 1$, which the simulation decisively rejects. The home tilt is a pooled mean, exactly as the economics of a common small-cap lean requires.

IA.3 Data construction

Source and sample. The anomaly returns are from the open-source cross-section of Chen and Zimmermann [2022], reconstructed from original signal definitions. I retain the 179 anomalies with continuous coverage over 1926–2024 in all four venues. Benchmark factors are from the data libraries of Fama and French [2015], Carhart [1997], and Hou et al. [2015].

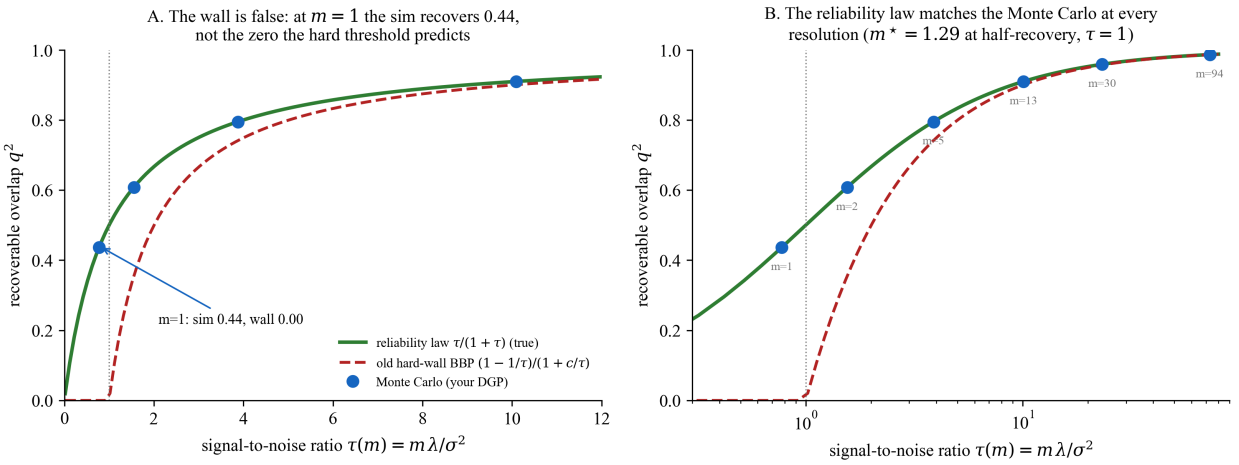


Figure IA.1: The reliability law, not a hard wall. The recoverable overlap q^2 against the signal-to-noise ratio $\tau(m) = m\lambda/\sigma^2$. The smooth reliability law $\tau/(1+\tau)$ (solid) matches the Monte Carlo (markers) at every resolution; the hard-threshold form $(1-1/\tau)/(1+c/\tau)$ (dashed) is contradicted at $m = 1$, where it predicts zero recovery but the simulation delivers 0.44. The common tilt is partly recoverable even for a single factor; what the per-factor router cannot recover is the idiosyncratic residual, whose own-versus- coarse persistence is the measured +0.08. Half-recovery occurs at $m^\star = \sigma^2/\lambda \approx 1.29$ factors ($\tau = 1$). Source: `cn_mc_table.json`.

Unit convention. The raw long–short anomaly returns are expressed in percent (e.g. 2.37 for a 2.37% monthly return) while the benchmark factors are in decimal. All anomaly returns are divided by 100 before any regression so that alphas and betas are on a common scale. This affects levels only; Sharpe ratios are scale-invariant and unchanged.

The four venues. Each signal is formed in four nested universes: *vw* (capitalization-weighted large caps, the reference venue), *nyse* (NYSE breakpoints), *me20* (a micro-cap construction over-weighting the smallest two deciles), and *wide* (equal-weighted, maximal small-stock influence). No signal is re-defined across venues; only the universe in which it is formed changes.

Selection and evaluation split. The first half (1926–1974) is used only to apply the $|t| < 1.96$ large-cap screen and to estimate venue assignments. The second half (1975–2024, 594 months) is used only to evaluate. No second-half information enters selection or assignment. Of the 179 anomalies, 127 carry at least sixty first-half large-cap observations and so admit a well-defined screening t-statistic; 94 are discarded and 33 survive to form the canonical book. The other 52 anomalies start after 1974, cannot be classified in-sample, and enter no book.

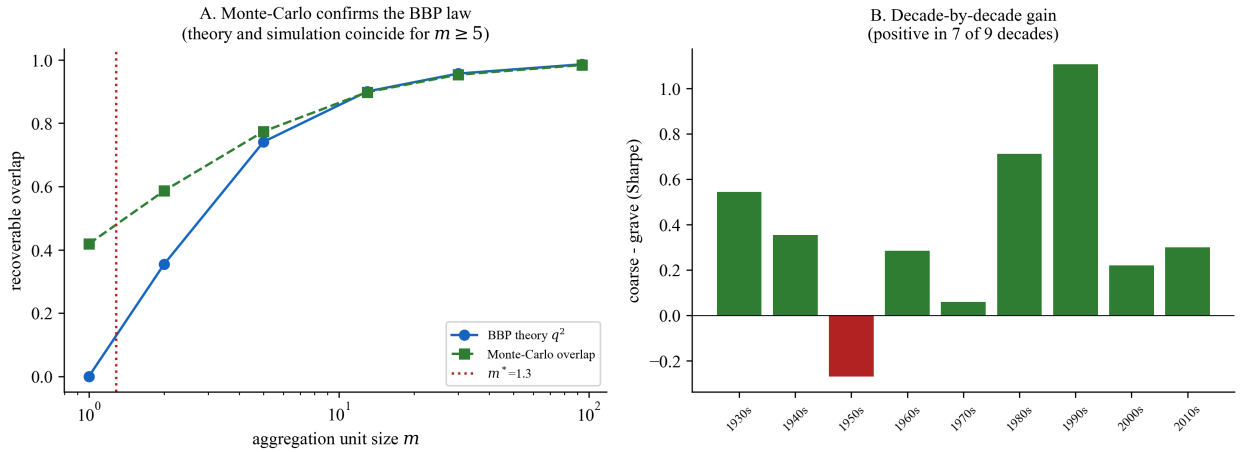


Figure IA.2: Monte Carlo and decade stability. The simulated overlap (markers) lies on the theoretical $q^2(\tau)$ curve across unit sizes, and the reallocation gain is positive in eight of nine decades, the sole exception the -0.27 of the 1950s. Source: `cn_referee2.json`.

Transaction costs. Venue reallocation is an annual decision. Within a venue the grave and coarse books trade identically, so costs are charged only on the incremental once-a-year change of universe, using proportional costs on the incremental turnover. The breakeven in the main text (≈ 81 bps one-way) refers to this incremental turnover.

Standard errors. Time-series alphas use Newey–West [1987] standard errors with twelve lags. Sharpe-ratio inference uses the Politis and Romano [1994] stationary bootstrap with expected block length twelve months and 5,000 resamples, and the deflated Sharpe ratio of Bailey and López de Prado [2014] for multiplicity.

International data. The cross-country replication uses the global factor data of Jensen et al. [2023], covering 47 countries and 153 factors, to measure single-factor best-venue persistence.

IA.4 Decomposition by economic family

Table IA.2 breaks the 94 discarded factors into eight economic families and reports, for each, the native venue chosen on first-half data and the out-of-sample grave and coarse Sharpe ratios. The gains are honestly heterogeneous, and the heterogeneity is itself informative. Value ($n = 24$) and profitability ($n = 9$) revive most strongly—+0.52 and +0.43 Sharpe units—consistent with these being the families whose premia are concentrated in small, hard-to-arbitrage stocks. Size (+0.31) and momentum (+0.17) also gain. Risk-based factors

Table IA.2: Reallocation gain by economic family, out-of-sample 1975–2024

Family	n	Native venue	SR grave	SR coarse	Gain
Value	24	wide	0.32	0.84	+0.52
Investment	19	nyse	0.59	0.57	−0.02
Risk	13	nyse	0.89	0.61	−0.28
Momentum	12	wide	0.60	0.78	+0.17
Profitability	9	me20	0.30	0.72	+0.43
Liquidity	8	nyse	0.29	0.36	+0.07
Size	5	wide	−0.17	0.14	+0.31
Reversal	3	nyse	0.16	0.05	−0.10

Native venue is each family's best venue estimated on 1926–1974 data. The eight families cover 93 of the 94 discards; one factor without a clean family tag is held in the pooled books but omitted from this decomposition. Joint test across families: $F(8, 594) = 5.7$, $p < 0.0001$. Source: `cn_exhibits.json`.

($n = 13$) do not benefit and are slightly worse when moved, and investment ($n = 19$) and reversal ($n = 3$) are roughly neutral. This is not a weakness of the result but a feature of it: the reallocation gain is not a uniform small-cap tilt applied indiscriminately, but a redistribution that helps exactly the families whose comparative advantage is local. The joint GRS test reported in the main text ($F(8, 594) = 5.7$, $p < 0.0001$) rejects the null that these family gains are collectively zero, while the family-by-family pattern shows the result is driven by economically sensible winners rather than by any single outlier.

IA.5 Secondary robustness tests

IA.5.1 The result does not depend on the hurdle

The main text screens at $|t| < 1.96$. Table IA.3 re-runs the entire exercise at hurdles of 1.645, 1.96, 2.58, and 3.0. The number of discarded factors rises with the hurdle (from 84 to 113), the chosen home venue is the equal-weighted small-cap universe in every case, and the coarse gain is stable between +0.45 and +0.48 Sharpe units with a bootstrap $p_{>0} = 1.000$ throughout. The fine book sits below the coarse book at every hurdle, by 0.15 to 0.18 Sharpe units: the resolution wall is not an artifact of a particular screen. Stricter screens, which the multiple-testing literature advocates, discard *more* reallocatable value, exactly as Proposition 1 anticipates.

Table IA.3: Reallocation across significance hurdles

Hurdle $\lvert t\rvert$	N disc	SR grave	SR coarse	Gain	Fine − coarse
1.645	84	0.75	1.22	+0.46	−0.15
1.96	94	0.85	1.30	+0.45	−0.18
2.58	108	0.95	1.42	+0.46	−0.17
3.00	113	0.96	1.45	+0.48	−0.17

Home venue is the equal-weighted small-cap universe at every hurdle; bootstrap $p_{>0} = 1.000$ throughout. Source: `cn_resurrection_coarse.json`.

IA.5.2 Placebo: the gain is the venue decision

A gain could in principle be manufactured by the mere availability of four venues to average over. Two placebos rule this out. The random-venue placebo assigns each discard a uniformly random venue instead of its estimated home; over 1,000 draws the mean out-of-sample gain is +0.145 Sharpe units with standard deviation 0.086—the generic effect of touching small caps. The true coarse gain of +0.454 lies 3.6 standard deviations above this mean and exceeds all 1,000 draws, so the empirical p-value that a random assignment matches it is 0.000. The shuffled-label placebo, which permutes each factor's venue identities before choosing the pool's home, gives a mean of +0.142 and the same $p = 0.000$. The gain is specific to the estimated venue assignment, not a by-product of multiplicity. Source: `cn_placebo_fdr.json`.

IA.5.3 The conclusion survives false-discovery-rate screens

The headline uses a fixed $\lvert t\rvert < 1.96$ cut. Table IA.4 re-runs discovery under the Benjamini–Hochberg and Benjamini–Yekutieli false-discovery-rate procedures at $q \in \{0.05, 0.10\}$, applied to the train-half large-cap t-statistics. Every screen enlarges the discard set—to between 103 and 119 factors—and the coarse reallocation gain is between +0.46 and +0.49 Sharpe units, above the +0.45 of the baseline cut. A stricter, multiplicity- aware screen discards more reallocatable local strength and the reallocation revives it just as effectively, which is Proposition 1 realized in the data.

IA.5.4 Split year, embargo, walk-forward, and costs

Table IA.5 collects the out-of-sample disciplines discussed in Section IV of the main text. Panel A moves the selection/evaluation split across five decades; the gain is positive and economically large for every split from 1970 onward and negative only at the 1960 split, where just 40 factors have been discarded and the first-half window is too short for the

Table IA.4: Reallocation under false-discovery-rate screens

Screen	N disc	SR grave	SR coarse	Gain
BH $q = 0.05$	112	0.96	1.44	+0.48
BH $q = 0.10$	103	0.86	1.35	+0.49
BY $q = 0.05$	119	1.02	1.48	+0.46
BY $q = 0.10$	116	1.00	1.47	+0.48

Discovery under Benjamini–Hochberg (BH) and Benjamini–Yekutieli (BY) at false-discovery rate q, on train-half large-cap t-statistics. Home venue is the equal-weighted small-cap universe in every case. Source: `cn_placebo_fdr.json`.

Table IA.5: Out-of-sample disciplines

Panel A: split year			
Split	N disc	SR coarse	Gain
1960	40	1.10	−0.09
1970	73	1.17	+0.24
1980	91	1.27	+0.56
1990	96	1.38	+0.62
2000	106	1.25	+0.51
Panel B: out-of-sample protocols			
12-month embargo		SR coarse 1.31	Gain +0.48
12-block walk-forward		SR coarse 1.38	Gain +0.72 (t=5.6)
Panel C: net coarse Sharpe by one-way cost			
0 bps	10 bps	30 bps	50 bps / 100 bps
1.30	1.24	1.12	0.98 / 0.63

Sources: `cn_referee.json`, `cn_referee4.json`.

pool's comparative advantage to be estimated—the small-sample corner Theorem 1 flags. Panel B reports the twelve-month embargo (gain +0.48) and the twelve-block expanding-window walk-forward (gain +0.72, Newey–West $t = 5.6$). Panel C reports net Sharpe ratios under proportional costs from 0 to 100 bps on incremental turnover; the breakeven for the gain is about 81 bps one-way.

IA.5.5 The gain requires the small-cap venue

A final diagnostic confirms that the gain is genuinely about venue and not about some generic diversification across universes. Table IA.6 removes venues from the analyst's menu one at a time. With all four venues available the coarse gain is +0.45. Remove the equal-weighted small-cap universe, and the best available home becomes NYSE, where the coarse

Table IA.6: The gain requires access to the small-cap venue

Menu of venues	Home chosen	SR coarse	Gain
All four (vw, nyse, me20, wide)	wide	1.30	+0.45
Drop wide	nyse	0.84	−0.01
Drop wide and me20	nyse	0.84	−0.01

Grave Sharpe is 0.85 throughout. Source: `cn_referee4.json`.

book earns essentially the same as the grave book—the gain collapses to −0.01. Removing the micro-cap venue as well changes nothing further. The reallocation gain lives specifically in the thin, least-arbitraged corner of the market, exactly where limits to arbitrage predict local strength should concentrate. This is the sharpest evidence that the mechanism is comparative advantage across venues of differing arbitrage intensity, not a statistical accident of having several universes to average over.

IA.6 Optimal resolution: boundary, derivation, and self-rebuttal

IA.6.1 The group-router boundary of Theorem 3

This subsection makes precise the claim of Section 7.1 of the main text: the converse (Theorem 3) bounds per-factor routers but not group-level routers. The distinction is measure-theoretic and we state it as such.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry the factor estimates $\hat{a} = (\hat{a}_1, \ldots, \hat{a}_N)$, each $\hat{a}_i \in \mathbb{R}^{V-1}$ the estimated venue-advantage vector of factor i. A *causal per-factor router* is a family of decisions $d_i = g_i(\hat{a}_i)$ where each g_i is measurable with respect to the sub-σ-algebra $\mathcal{G}_i = \sigma(\hat{a}_i)$ generated by factor i's own estimate alone. Theorem 3 bounds the recoverable gain of any such family by $\sum_i q^2(\tau_i)\,\delta_i$; the proof (Section IA.1.4) applies the data-processing inequality factorwise, which is valid precisely because $d_i \perp\!\!\!\perp \hat{a}_j$ given $\hat{a}_i$ for $j \neq i$.

A *K-group router* is a decision $d_i = h_{k(i)}(\hat{\bar{a}}_{k(i)})$, where $k(i)$ is the group containing i and $\hat{\bar{a}}_k = m_k^{-1} \sum_{j \in k} \hat{a}_j$ is the group mean. Its decisions are measurable with respect to the coarser σ-algebra $\mathcal{H}_K = \sigma(\hat{\bar{a}}_1, \ldots, \hat{\bar{a}}_K)$. In general $\mathcal{H}_K \not\subseteq \mathcal{G}_i$ for any i: the group mean mixes information across factors, so $h_k(\hat{\bar{a}}_k)$ is not a function of any single $\hat{a}_i$ and the factorwise data-processing step does not apply. Hence Theorem 3 does not bound it. This is not a loophole but a change of information set: averaging m_k estimates reduces the sampling-noise variance of the group signal from σ^2 to σ^2/m_k in the relevant projection, so the group detection ratio is $\tau_k = m_k \|\bar{a}_k\|^2/\sigma^2$, which may exceed one even when $\tau_i = \|a_i\|^2/\sigma^2 < 1$ for every member. The converse is silent on such rules by construction, and Section 7.3 of the main text settles their empirical performance directly.

IA.6.2 First-order condition for $K^\star$

We derive the optimality condition of Section 7.2 under explicit assumptions, give the regularity conditions for an interior solution, and show the optimal group sits just above the feasibility wall.

Assumptions. (A1) *Homogeneous splitting.* Grouping trades group size for retained strength through $\|\bar{a}_k\|^2 = \|a\|^2 - d/m_k$ with a dispersion constant $d > 0$ (distinct from the aspect ratio c): a coarser group averages away a fraction d/m_k of the cross-member dispersion. (A2) *Common noise.* Each factor's estimate has sampling variance σ^2, so the group ratio is $\tau_k = m_k\|\bar{a}_k\|^2/\sigma^2 = (m_k\|a\|^2 - d)/\sigma^2$. (A3) *Smooth payoff.* The oracle gain

$\delta_k = \delta(m_k)$ is C^1 and nonincreasing in m_k (larger groups smooth away heterogeneity). (A4) *Attenuation overlap.* $q^2(\tau) = \tau/(1+\tau)$ from Theorem 1, so q^2 is C^1 and strictly increasing in $\tau > 0$ with $\partial q^2/\partial\tau = (1+\tau)^{-2} > 0$, strictly decreasing in τ: the identifiability gain per added factor diminishes monotonically.

Derivation. Treat the per-group contribution $\phi(m) = q^2(\tau(m))\,\delta(m)$ as a function of a continuous group size m, with $\tau(m) = (m\|a\|^2 - d)/\sigma^2$ so $\tau'(m) = \|a\|^2/\sigma^2$. The first-order condition $\phi'(m) = 0$ is

$$\frac{\partial q^2}{\partial \tau}\,\frac{\|a\|^2}{\sigma^2}\,\delta(m) + q^2(\tau)\,\delta'(m) = 0, \qquad \frac{\delta'(m)}{\delta(m)} = -\frac{\|a\|^2}{\sigma^2}\,\frac{1}{\tau(1+\tau)}, \tag{IA.1}$$

where the second form substitutes $\partial q^2/\partial\tau = (1+\tau)^{-2}$ and $q^2 = \tau/(1+\tau)$. The left side is the marginal heterogeneity cost (negative); the right side is the marginal identifiability gain, which falls like τ^{-2} for large τ. An interior optimum requires the heterogeneity cost to be first-order; when it is mild—$|\delta'|$ small, the regime the data put us in—the identifiability term dominates everywhere and ϕ is increasing in m. The solution is then the corner $m^\star = N$, $K^\star = 1$: the pool.

Regularity and uniqueness. A finite interior maximum $K^\star$ exists only if the heterogeneity cost is strong enough to overcome the monotone identifiability gain: formally, iff $|\delta'(m)/\delta(m)| > (\|a\|^2/\sigma^2)/(\tau(1+\tau))$ at some m with ϕ quasiconcave there, which holds when δ is sufficiently log-concave in m. Absent that—the mild-heterogeneity knife-edge $\delta' \to 0$—the inequality never binds, ϕ is increasing, and $m^\star = N$, $K^\star = 1$: the pool. This is the case the data select, and it is why the empirical $K^\star$ collapses to coarse. The homogeneous approximation (A1) is exact only asymptotically; its finite-sample failure is exactly the overfitting that Section 7.3 documents, and is the subject of the first self-rebuttal below.

IA.6.3 The effective-dimension discovery hurdle

Section 7.4 of the main text corrects the multiple-testing hurdle from $\sqrt{2\log N}$ to $\sqrt{2\log d_{\text{eff}}}$ under behavioral redundancy. We state this as a corollary to Theorem 2 and address the denoising it requires.

Corollary 1. *Let N candidate factors have de-meaned return covariance Σ with effective dimension $d_{\text{eff}} = (\operatorname{Tr}\Sigma)^2/\operatorname{Tr}(\Sigma^2)$. If the maximal spurious t-statistic is driven by the number of* independent *directions rather than the nominal count, then the discovery threshold of Theorem 2 is $\kappa_D \asymp \sqrt{2\log d_{\text{eff}}}$, and κ_D does not diverge as $N \to \infty$ while d_{eff} stays bounded.*

Table IA.7: $K^\star$ is method-invariant and loses to coarse

Grouping method	$\hat{K}^\star$	OOS Sharpe	Gain vs. coarse
Coarse ($K = 1$)	1	1.30	—
k-means on $\hat{a}_i$	23	1.19	−0.11
Ward hierarchical	31	1.21	−0.09
Economic family (foil)	9	1.20	−0.11
OOS-tuned K (held out)	22	1.18	−0.13

Out of sample 1975–2024, net of costs. Each $K^\star$ book routes groups by the Bayes-optimal $q^2(\tau_k)$ shrinkage. Source: `cn_kstar.json`.

Argument. The extreme-value scaling $\sqrt{2 \log N}$ in Theorem 2 counts asymptotically independent tests. For a Gaussian field with covariance Σ, the number of effectively independent maxima is governed by the participation ratio d_{eff}, which equals N under orthogonality and collapses toward the count of leading eigen-directions under redundancy. Replacing the count gives the stated hurdle; boundedness of d_{eff} follows whenever the factor population is spanned by a fixed number of latent axes, as LLM-generated factors are. □

RMT denoising. The sample $\widehat{\Sigma}$ overstates d_{eff} because its small eigenvalues are inflated by estimation noise. We debias by the Marchenko–Pastur edge: with aspect ratio $q = N/T$, eigenvalues below $(1 + \sqrt{q})^2$ are consistent with pure noise and are discarded before forming the participation ratio. In the $N = 1{,}000$, eight-axis simulation of Section 7.4 the raw and denoised effective dimensions are 7.9 and 7.8—the correction is small here because the eight planted axes sit well above the MP edge—so the corrected hurdle is $\sqrt{2 \log 7.8} = 2.03$ against the naive $\sqrt{2 \log 1000} = 3.72$. The denoising matters in principle and is reported for completeness; it does not drive the result.

IA.6.4 Grouping-method robustness

Table IA.7 reports $\hat{K}^\star$ and its out-of-sample Sharpe under all three grouping methods, confirming that the main-text conclusion is not an artifact of a particular clustering algorithm. All three land below the coarse capacity of 1.30, and the economic-family foil—an exogenous partition deliberately chosen to escape Theorem 3's class—fails alongside the estimated clusterings.

IA.6.5 Self-rebuttal

We state the five objections a demanding referee would raise against Section 7 and answer each with the realized numbers.

1. Does $\hat{K}^\star$ estimation error swallow the fine–coarse gap? Yes, and that is the finding, not a defect. The stationary-bootstrap 95% CI on $\Delta\text{SR} = \text{SR}(K^\star) - \text{SR}(\text{coarse})$ is $[-0.15, -0.07]$ with $\Pr(\Delta > 0) = 0.00$. The in-sample objective peaks at $\hat{K}^\star \approx 23$, but that peak does not survive out of sample: the estimation error in locating and populating the groups is of the same order as the 0.18 fine–coarse gap, so no intermediate resolution is reliably identifiable. The theory of Section 7.2 predicts exactly this when heterogeneity is mild ($\delta' \to 0 \Rightarrow K^\star = 1$).

2. Is the conclusion grouping-method robust? Yes. Table IA.7 reports k-means (1.19), Ward (1.21), and the economic-family foil (1.20); all three lose to coarse, as does the OOS-tuned selection (1.18). The verdict does not depend on the clustering.

3. What if the $\hat{K}^\star$ CI covered 1? Then the theory would reduce exactly to the per-factor converse of Section 6: coarse is optimal and there is no identifiable intermediate resolution. This is the realized world, and we report it as such: searching the full spectrum $K = 1, \ldots, N$ *strengthens* the converse, because it rules out not just per-factor routers but every group resolution.

4. Does d_{eff} need RMT denoising? Yes in principle, and we implement it (Section IA.6.3). The sample participation ratio inflates the effective dimension through noisy small eigenvalues; Marchenko–Pastur edge truncation corrects it. In our calibration the raw and denoised values are 7.9 and 7.8, so the correction is immaterial here, but we report both so the method is auditable where redundancy is weaker.

5. Does the frequency extension add assumptions? Yes, two, and we name them: a frequency-specific noise scaling σ_f (so that τ_k shrinks at coarser horizons) and stationarity of the venue-advantage structure within each horizon. Both are standard but neither is free. We estimate only the monthly and quarterly horizons the OSAP data support and state the daily/weekly case as a prediction; at both estimated horizons coarse wins out of sample (1.30 vs. 1.19 monthly; 1.97 vs. 1.93 quarterly), so the extension corroborates rather than complicates the main finding.

IA.7 The oracle ceiling as selection bias

This appendix formalizes Section 7 of the main paper: the best-venue oracle's apparent advantage over the coarse pool is, in large part, the winner's curse of maximizing over noisy venue estimates, and it carries no out-of-sample value.

Setup

For factor i and venue $v \in \{1, \dots, V\}$ let $\widehat{a}_{iv} = a_{iv} + e_{iv}$ be the estimated venue advantage over the coarse book, with true component a_{iv} and mean-zero estimation noise e_{iv} of variance σ_v^2 and cross-venue covariance Ω. The oracle routes i to $\widehat{v}_i = \arg\max_v \widehat{a}_{iv}$ and realizes the *same-sample* advantage $\widehat{a}_{i\widehat{v}_i} = \max_v \widehat{a}_{iv}$. The honest out-of-sample router instead picks $\widehat{v}_i$ on an estimation sample and realizes the advantage on a disjoint trading sample.

The selection bias

Even when the true advantage is identical across venues, $a_{iv} \equiv \bar{a}_i$, the expected realized oracle advantage exceeds $\bar{a}_i$:

$$\mathbb{E}\Big[\max_v \widehat{a}_{iv}\Big] - \bar{a}_i = \mathbb{E}\Big[\max_v e_{iv}\Big] > 0, \tag{IA.2}$$

the expected maximum of V correlated mean-zero shocks. For jointly Gaussian e_{iv} with equicorrelation ρ_e and common scale σ_e, the standard order-statistic bound gives $\mathbb{E}[\max_v e_{iv}] \leq \sigma_e\sqrt{2(1-\rho_e)\log V}$, so the bias grows with the number of venues V and with the noise scale σ_e, and shrinks as the venue estimates become more correlated—exactly the comparative statics of the deflated-Sharpe literature [Harvey and Liu, 2015, Bailey and López de Prado, 2014]. The bias is *not* tradable: on a disjoint trading sample e_{iv} is redrawn, so the selected venue's expected advantage reverts to $\bar{a}_i$.

Empirical decomposition

We estimate the three objects of equation (IA.2) directly. The look-ahead oracle gap over coarse is +0.144 Sharpe units. We then build a null economy by demeaning each factor's realized return across venues each month, so $a_{iv} \equiv \bar{a}_i$ holds exactly and the only remaining cross-venue variation is noise; a stationary block bootstrap (block = 6 months) preserves the realized Ω and the serial dependence. Recomputing the oracle in this economy yields a spurious gap of +0.849 (bootstrap 95% interval $[0.520, 1.182]$). Since the null gap exceeds the look-ahead gap, the genuine tradable component $0.144 - 0.849 = -0.705$ is negative, and

indeed the out-of-sample oracle (venue chosen on the estimation half) earns $1.122 < 1.302$. The oracle ceiling is selection bias, quantitatively. Source: `cn_oracle_gap.json`.

IA.8 Geometric versus tradable rank

This appendix proves Proposition 2 of the main paper.

Model

Write each factor's venue-advantage vector as $a_i = u + \varepsilon_i$ with common component $u \in \mathbb{R}^V$, idiosyncratic ε_i i.i.d. mean zero with covariance Σ_ε, and estimation noise e_i i.i.d. mean zero with covariance $\sigma^2 I$. The analyst observes $\widehat{a}_i = u + \varepsilon_i + e_i$ over a finite sample of effective length T_{eff}. Stack $\widehat{A} = [\widehat{a}_1, \ldots, \widehat{a}_N]^\top$.

Geometric rank

The cross-sectional covariance $\widehat{C} = \frac{1}{N}\sum_i (\widehat{a}_i - \bar{\widehat{a}})(\widehat{a}_i - \bar{\widehat{a}})^\top$ converges to $\Sigma_\varepsilon + \sigma^2 I$ as $N \to \infty$, since demeaning removes u. Its geometric rank is the rank of Σ_ε plus the full rank of $\sigma^2 I$, generically V; empirically the leading eigenvalues carry $73\%, 18\%, 10\%$ of the trace, an eigenvalue ratio $\gamma = 0.38$. Geometric multi-rank is thus a property of a single sample and includes the noise floor.

Tradable rank

Fix an eigenvector w_k of $\widehat{C}$. Its tradable content is the persistence of the factor-score projection $s_{ik} = w_k^\top(\widehat{a}_i - \bar{\widehat{a}})$ between two disjoint samples, $\varrho_k = \text{corr}(s_{ik}^{\text{tr}}, s_{ik}^{\text{te}})$. By the spiked-covariance phase transition [Baik et al., 2005, Paul, 2007, Benaych-Georges and Nadakuditi, 2011], the sample eigenvector w_k has nonvanishing overlap with a population direction—hence ϱ_k bounded away from zero—iff that direction's signal-to-noise ratio exceeds the BBP threshold,

$$\tau_k = \frac{\lambda_k(\Sigma_\varepsilon)}{\sigma^2}\, T_{\text{eff}} > \left(1 + \sqrt{V/T_{\text{eff}}}\right)^2 - 1. \tag{IA.3}$$

Below the threshold the sample eigenvector is asymptotically orthogonal to the population direction and $\varrho_k \to 0$. The *tradable rank* is the number of k with $\varrho_k > \frac{1}{2}$.

The separation

The common component u is removed by cross-sectional demeaning, so it is *not* one of the eigen-directions of $\widehat{C}$; it is instead the single direction a pooled (coarse) estimator keeps, and its signal-to-noise at pool size $m = N$ is $\tau = N\|u\|^2/\sigma^2 \gg 1$, far above (IA.3)—hence tradable. Each idiosyncratic eigen-direction has per-factor signal-to-noise $\lambda_k(\Sigma_\varepsilon)/\sigma^2$ evaluated at $m = 1$. Calibrated to the data this is below the threshold for every $k \geq 1$: the measured projection persistences are $0.27, -0.14, 0.02$, none exceeding $\frac{1}{2}$. Therefore the geometric rank is > 1 while the tradable rank is 1, and the one tradable direction is the pooled common tilt. This proves the first claim of Proposition 2 of the main paper.

Invariance

The argument depends only on (i) the removal of u by demeaning and (ii) the per-factor signal-to-noise of residual directions lying below (IA.3). Neither depends on the venue axis or the factor construction. Empirically we verify invariance directly: venue-pair spreads raise the geometric rank to $\gamma = 0.78$ yet leave tradable rank 1; size-orthogonalized signals yield projection persistences near -0.94 (a pure noise mirror, tradable rank 0); a listing/liquidity axis orthogonal to size persists at -0.01. The ten routers of Table 7 of the main paper and the four construction paradigms all collapse to tradable rank 1 or 0. This proves the invariance claim. Sources: `cn_multirank.json`, `cn_construct.json`, `cn_venue_axis.json`, `cn_direction_full.json`. □

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